Andhra University College of Engineering

Electrical and Electronics Engineering | Course Code: EE301

Power Systems Engineering - Power Systems Transmission Lines & Fault Analys

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Overview: Flux linkages, GMR and GMD calculation, Ferranti effect, Symmetrical fault analysis,

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Transmission Line Modeling
Inductance and capacitance of single and 3-phase lines, GMD and GMR formulations, Skin and Proximity

Unit 2: Performance of Lines & Voltage Control
Short, medium Nominal T and Pi, and long lines, ABCD parameters, Surge impedance loading, Ferranti e

Unit 3: Symmetrical & Unsymmetrical Faults
Symmetrical fault calculations, Symmetrical components Zero, Positive, Negative sequences, LG, LL, L

Unit 4: Power Flow Analysis
Bus classification Slack, PQ, PV, Gauss-Seidel and Newton-Raphson power flow formulation

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Derive ABCD parameters for a Nominal-Pi transmission line network and plot receiving end circle diagra
2. Formulate sequence impedance networks for a single line-to-ground LG fault and determine subtransient
3. Explain the algorithm for Newton-Raphson load flow in polar coordinates with Jacobian matrix elements.

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* I.J. Nagrath, D.P. Kothari - Modern Power System Analysis McGraw-Hill
* Hadi Saadat - Power System Analysis PSA Publishing

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